PR 4 Probabilitas dan Statistika 13524044/Narendra Sharma

8.25 The average life of a bread-making machine is 7 years, with a standard deviation of 1 years. Assuming that the lives of these machines follow approximately a normal distribution, find

a) probability that the mean life of a random sample of 9 such machines falls between 6.4 and 7.2 years;

b) value of x to the right of which 15% of the means computed from random sample of size 9.

Jawab:

a) $P(6.4 < x < 7.2)$

$$z_1 = \frac{\bar{X}_1 - \mu}{\sigma/\sqrt{n}} = \frac{6.4 - 7}{1/\sqrt{9}} = \frac{-0.6}{1/3} = -0.6 \times 3 = -1.8$$

$$z_2 = \frac{\bar{X}_2 - \mu}{\sigma/\sqrt{n}} = \frac{7.2 - 7}{1/\sqrt{9}} = \frac{0.2}{1/3} = 0.2 \times 3 = 0.6$$

$$P(-1.8 < z < 0.6) = P(z < 0.6) - P(z < -1.8) = 0.7257 - 0.0359$$

$$= \underline{\underline{0.6814}}$$

b) $P(\bar{X} > x) = 0.15$

$$P(\bar{X} > x) = 0.15 \Rightarrow P(\bar{X} < x) = 1 - 0.15 = 0.85$$

$$P(z < z) = 0.85 \rightarrow z \approx 1.04$$

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \Rightarrow \bar{X} = \mu + z\left(\frac{\sigma}{\sqrt{n}}\right)$$

$$= 7 + 1.04\left(\frac{1}{3}\right)$$

$$\approx 7.3467$$

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8.31. Consider Case Study 8.2. Suppose 18 specimens were used for each type of paint in an experiment and $\bar{y}_A - \bar{y}_B$ is the actual difference in mean drying time, turned out to be 1.0.

a) reasonable if the two population mean drying times truly are equal?

b) If someone did the experiment 10.000 times under the condition that $\mu_A = \mu_B$, in how of those 10.000 would there be a difference $\bar{y}_A - \bar{y}_B > 1.0$?

Jawab:

a) $$z = \frac{(\bar{X}_A - \bar{X}_B) - (\mu_A - \mu_B)}{\sigma_{\bar{X}_A - \bar{X}_B}} = \frac{1.0 - 0}{\sqrt{1/9}} = 3.0$$

$$P(z \geq 3.0) = 1 - P(z < 3.0)$$

$$= 1 - 0.9987 = 0.0013$$

$$P(|\bar{X}_A - \bar{X}_B|) \geq 1.0 = P(z \leq -3.0) + P(z \geq 3.0)$$

$$= 0.0013 + 0.0013 = 0.0026 = 0.26\%$$

∴ tidak wajar jika diasumsikan kedua rata-rata populasi benar-benar sama (peluangnya rendah 0.26%)

b) $$P(\bar{X}_A - \bar{X}_B) \geq 1.0 = 0.0013$$

Jumlah eksperimen $= 10000 \times 0.0013$

$= 13$ //

8.49 A normal population with unknown variance has a mean of 20. Is one likely to obtain a random sample of size 9 from this population with a mean of 24 and a standard deviation of 4.1?

$v = n - 1 = 9 - 1 = 8$

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{24 - 20}{4.1/\sqrt{9}} = \frac{4}{4.1/3} \simeq 2.926$$

Dengan $v = 8$

$t_{0.01} = 2.896$

$t_{0.005} = 3.355$

nilai $t \geq 2.926$ berada di antara $t_{0.01}$ dan $t_{0.005}$

Nilai antara t dan $t_{0.01}$ hanya sedikit, sehingga dapat diproksimasi nilai t

$$0.05 < P(T \geq 2.926) < 0.01$$

aproksimasi : $0.05 < 0.009 < 0.01$

$P(|T \geq 2.926|) = 2 \times 0.009 = 0.018$ atau 1.8%

∴ peluang sangat rendah, maka asumsi ($\mu = 20$) tidak valid

9.27 Consider Study Case 9.1 with a larger sample of metal piece. The diameters are as follows: 1.01, 0.97, 1.03, 1.04, 0.99, 0.98, 1.01, 1.03, 0.99, 1.00, 1.00, 0.99, 0.98, 1.01, 1.02. Construct. Discuss how they are different and why.

a) 99% confidence interval on the mean diameter

b) 99% prediction interval on the next diameter

c) 99% tolerance interval of the central 95% distribution

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Jawab:

a) $\bar{x} = \frac{\sum x_i}{n} = \frac{1.01 + 0.97 + 1.03 + 1.04 + 0.99 + 0.98 + 1.01 + 1.03 + 0.99 + 1.00 + 1.03 + 0.99 + 0.98 + 1.01 + 1.02 + 0.94}{16}$

$= \frac{16.03}{16} = 1.001875$

$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \approx 0.0223257$

$v = n - 1 = 15$

$\gamma = 99\%, \alpha = 0.01$

$t_{\alpha/2} = t_{0.005}$ dengan $v = 15$

$t_{0.005} = 2.947$

$CI = \bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$

$= 1.001875 \pm 2.947 \cdot \frac{0.0223257}{\sqrt{16}}$

$= 1.001875 \pm 2.947 \cdot 0.005581$

$= 1.001875 \pm 0.01644 = (0.98544, 1.0183)$

$CI = (0.9854, 1.0183)$

b) $PI = \bar{x} \pm t_{\alpha/2} \cdot s \sqrt{1 + \frac{1}{n}}$

$= 1.001875 \pm 2.947 \times 0.0223257 \times \sqrt{1.0625}$

$= 1.001875 \pm 0.06786$

$= (0.93400, 1.0697)$

$PI = (0.9340, 1.0697)$

c) $TI = \bar{x} \pm ks$

$k_{n=16, \gamma=0.95, 1-\alpha=0.99} = 3.018$

$TI = 1.001875 \pm 3.018 \times 0.0223257$

$= 1.001875 \pm 0.06739$

$TI = (0.9345, 1.0692)$

Semua interval pada soal lebih sempit dibandingkan interval pada Case Study, Dikarenakan naik dan turun sehingga $(s/\sqrt{n})$ juga berkurang, sehingga interval lebih sempit

9.49 Two different brands of Latex paint are being considered for use. Fifteen specimens each of type of paint were selected drying times:

Paint A	Paint B
3.5 2.7 2.9 4.7 3.6	4.7 3.9 4.5 5.3 4.0
3.7 3.3 5.2 4.2 2.9	5.3 4.3 6.0 5.2 3.7
4.4 5.2 4.6 4.1 3.4	5.5 6.2 5.1 5.4 4.8
$\bar{x} = 3.82$ $s^2 = 0.6214$	$\bar{x} = 4.94$ $s^2 = 0.6676$

$\sigma_A = \sigma_B$. 95% CI on $\mu_B - \mu_A$, where μ_A and μ_B are the mean drying times.

Jawab:

$S_p^2 = \frac{(n_A - 1) s_A^2 + (n_B - 1) s_B^2}{n_A + n_B - 2} = \frac{(15-1)(0.6214) + (15-1)(0.6676)}{28}$

$= \frac{14(0.6214) + 14(0.6676)}{28} = \frac{14 \cdot (0.6214 + 0.6676)}{28} = \frac{1.289}{2} = 0.6$

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$S_p = \sqrt{0{,}6492} = 0{,}80125$

$\alpha = 0{,}05 \Rightarrow \alpha/2 = 0{,}025$

$V = 28$

$t = 0{,}024$ dengan $V = 28$ adalah $2{,}048$

$ME = t_{0/2,V} \cdot S_p \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}$

$ME = 2{,}048 \times 0{,}80125 \sqrt{\frac{1}{15} + \frac{1}{15}}$

$= 2{,}048 \cdot 0{,}80125 \cdot \sqrt{\frac{2}{15}}$

$= 2{,}048 \cdot 0{,}80125 \sqrt{0{,}1333} = 0{,}5989$

$CI = (\bar{X}_B - \bar{X}_A) \pm ME$

$= 4{,}94 - 3{,}82 \pm 0{,}5989$

$= 1{,}12 \pm 0{,}5989$

$= 0{,}5211 \cap 1{,}7189$

$CI = (0{,}5211 , 1{,}7189)$

9.53 a) A random sample of 200 voters in a town is selected and 114 are found to support an democratic sub. Find the 96% CI

b) What can assert with 96% confidence about the possible size of our error if we estimate the fraction of voters sub to be 0.57

Jawab:

$\hat{p} = \frac{x}{n} = \frac{114}{200} = 0{,}57$

$\hat{q} = 1 - \hat{p} = 1 - 0{,}57 = 0{,}43$

$n\hat{p} = 200 \cdot 0{,}57 = 114 \geq 5$ ✓

$n\hat{q} = 200 (0{,}43) = 86 \geq 5$ ✓

normal ✓

untuk 96%, $\alpha = 0{,}04 \Rightarrow \alpha/2 = 0{,}02$

$1 - \alpha/2 = 1 - 0{,}02 = 0{,}98$

$Z_{0{,}02} \approx 2{,}05$

$S_{\hat{p}} = \sqrt{\frac{\hat{p}\hat{q}}{n}} = \sqrt{\frac{0{,}57 \cdot 0{,}43}{200}} = \sqrt{\frac{0{,}2451}{200}} \approx 0{,}035$

$CI = \hat{p} \pm Z_{\alpha/2} \cdot S_{\hat{p}}$

$= 0{,}57 \pm 2{,}05 \cdot 0{,}035$

$= 0{,}57 \pm 0{,}07176$

$= 0{,}4982 \cap 0{,}6418$

$CI = (0{,}4982 , 0{,}6418)$

b) Kesalahan yang mungkin terjadi menjadi 0,57 tidak akan melebihi ME atau $0{,}07176$ $(Z_{\alpha/2} \cdot S_{\hat{p}})$

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9.77 Construct a 99% CI for σ_1/σ_2 in 9.42, where σ_1 and σ_2 are respectively the standard deviation for the distances traveled / liter by VW and Toyota

Jawab:

$v = n_1 + n_2 - 2 = 20$

untuk tingkat kepercayaan 99% $\alpha = 0{,}10 \Rightarrow \alpha/2 = 0{,}05$

$t_{0{,}05,20} = 1{,}725$

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2} = \frac{(12-1)(1.0)^2 + (10-1)(0{,}8)^2}{20}$$

$$= \frac{11(1{,}0) + 9(0{,}64)}{20} = \frac{11 + 5.76}{20} = \frac{16{,}76}{20} = 0{,}83$$

$S_p = \sqrt{0{,}838} \approx 0{,}9154$

$$ME = t_{\alpha/2} \cdot S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$= 1{,}725 \times 0{,}9154 \cdot \sqrt{\frac{1}{12} + \frac{1}{10}}$$

$$= 1{,}725 \cdot 0{,}9154 \cdot \sqrt{0{,}183} \approx 0{,}678$$

$CI = (\bar{X}_1 - \bar{X}_2) \pm ME$

$= (16 - 11) \pm 0{,}678 = 5 \pm 0{,}678$

$= 4{,}322 \wedge 5{,}678$

$CI = (4{,}322 \; , 5{,}678)$

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